Thomas Nguyen
11888002
Professor Zhe Dang
Due 4/20/26 (extended to 4/21/26)

1.

For each node, create an array of k numbers to store the weights of the first k shortest paths from the initial node to that node.

- For the initial node: set the first entry to 0, all other entries to ∞
- For all other nodes: set all k entries to ∞
  Each phase updates the arrays at all nodes. Before a phase, let W_N be the array stored at node N. After the phase, the updated array is W'_N, computed as follows:
  For all nodes M where $M \to N$ is an edge in G:

1. Compute the multiset: $W_M[i] + Weight(M \to N)$ for $0 \leq i \leq k-1$
2. Also add the numbers already in W_N into this multiset
3. Select the k smallest elements from the multiset and store them as W'_N
   Repeat phases until no W'_N changes. Then reverse-traverse the graph from the final node back to the initial node using the stored arrays to recover the k shortest paths.
   Number of phases needed: kn, where n is the number of nodes in G. This can be improved by adding labels to nodes to indicate when they are finished and need no further updates.

2.

Use the original shortest path algorithm, but during path retrieval at the final state, avoid any path that contains a red edge immediately followed by a yellow edge.

3.

Build a bag that contains all color sequences from the initial node to the final node, forming an automaton. Run from the automaton's initial state (assuming the automaton has no loops, otherwise the count is infinite):

- For each node N, its count = the sum of the counts stored at all of its parent nodes
- The initial node stores 1
  The final answer is the count stored at the final node, which gives the total number of distinct color

sequences.

4.

Take the logarithm of all edge weights. Since $\log$ converts products into sums:

$$\log \prod w_i = \sum \log w_i$$

Run the standard shortest path algorithm on the log-transformed weights. The shortest path found corresponds to the minimum product path in the original graph.